Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 7

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Question 1

(i) How many new processes are created? Why?

- The loop runs 10 times
- Each fork() creates 1 new child process

Total new processes = 10

(ii) Which process executes function_A()?

Child process

Because,

else if (pid == 0)

- fork() returns 0 to the child
- So only the child enters this block

(iii) Whose PID is printed in Line A?

Child's PID (printed by parent)

Because,

- In parent → pid > 0
- pid contains child's process ID

(iv) Purpose of loop with wait() in Line B

To wait for all child processes to finish

Prevents,

- Zombie processes
- Parent exiting before children

Question 2

(i) For N = 5, how many processes are created?

Total processes = $2^5 = 32$

New processes = 31

(ii) Modified Program (ONLY parent creates 3 children)

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

void CPU() {
    printf("Child PID: %d is executing CPU()\n", getpid());
}

int main() {
    pid_t pid;
    int i;

    for(i = 0; i < 3; i++) {
        pid = fork();

        if(pid == 0) {  // child
            CPU();
            return 0;
        }
    }

    // parent waits
    for(i = 0; i < 3; i++) {
        wait(NULL);
    }

    return 0;
}
```

Question 3

Output of Line A,

PARENT: value= 40 (Because parent and child have separate memory copies)

Question 4

(i) Program A – how many processes?

Total = 2^4 = 16 processes

New processes = 15

(ii) Program B – output of Line A?

Value= 45

Value= 15

Question 5

1. Role of pthread_t tids[NUM_THREADS]

Stores thread IDs

2. Purpose of pthread_create() loop

To create 25 threads

3. Role of thread_routine()

Function executed by each thread (Prints thread number, Prints thread ID, Terminates thread using pthread_exit())

4. Why pthread_join() is used?

To make main thread wait for all threads